

Setup Guide

How to set up and run the Erdos solver pipeline on your own machine.

1. Requirements

- Python 3.9+
- A ChatGPT account (and optionally a DeepSeek account)

2. Install

```
git clone https://github.com/erichou1/erdos-open-problems.git
cd erdos-open-problems

# (recommended) create a virtual environment
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate

# install Python dependencies
pip install -r requirements.txt

# install the Chromium browser used by Playwright
playwright install chromium
```

3. Configure your ChatGPT Project URL

The pipeline opens new chats inside a ChatGPT Project. The URL is read from a local .env file (which is never committed).

```
cp .env.example .env
```

Then edit .env and set your own project URL:

```
CHATGPT_PROJECT_URL=https://chatgpt.com/g/g-p-<your-project-id>/project
```

To find it: open your project in ChatGPT and copy the URL from the address bar.

If you leave it unset, chats are created at plain <https://chatgpt.com>.

4. Log in (one time)

This opens a real browser window so you can sign in. Your session is saved in a local profile folder (git-ignored), so you only do this once.

```
python3 solve_submit.py --login
python3 deepseek_submit.py --login          # optional, for DeepSeek
```

5. (Optional) Fetch the problem set

The problems are already included under `erdos_problems/`. To re-download them:

```
python3 fetch_erdos.py
python3 fetch_categories.py
```

6. Run the pipeline

Submit a batch of problems (opens one chat per problem and sends the prompt):

```
python3 solve_submit.py --reverse --start 0 --limit 50
python3 deepseek_submit.py --reverse --start 0 --limit 50 --delay 60
```

Then collect answers and label each chat [solved]/[unsolved] + confidence:

```
python3 solve_rename.py --watch --interval 60
python3 deepseek_rename.py --watch --interval 45
```

Solutions are written to:

- erdos_problems/solutions/<category>/ (ChatGPT)
- erdos_problems/solutions_deepseek/<category>/ (DeepSeek)

A second, human-named copy of every answer is also written to an outputs/ folder, using the same name as the chat tab:

- outputs/chatgpt/<category>/Erdos #N [solved] 88%.md
- outputs/deepseek/<category>/Erdos #N [unsolved] 0%.md

7. Resuming after the computer is closed

Progress is saved to disk continuously, so you can close the computer (or stop a script) at any time and pick up where you left off:

- Submitted chats are recorded in the chat maps as they happen.
- Each answer is written to disk the moment it is collected.

Just re-run the same commands. Already-submitted problems and already-saved answers are skipped automatically, and the outputs/ copies are rebuilt from the saved solutions if they are missing. The --watch collectors also re-open the browser by themselves if the window is closed.

8. Privacy

These files stay local and are git-ignored:

- .env - your project URL
- .chatgpt_profile/, .deepseek_profile/ - browser sessions / cookies
- .chatgpt_chat_map.json, .deepseek_chat_map.json - conversation IDs
- *.log

See SETUP_GUIDE.pdf for a printable copy of this guide.